RUDRAKSH SYAL

Data Science Lead | Product & Decision Science

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PROFESSIONAL PROFILE

Data scientist with 10+ years translating ambiguous commercial and product questions into causal measurement, marketplace pricing, forecasting, optimization, and actionable decisions. Led production pricing systems and retail experiments, partnering with revenue management, product, engineering, and business leadership. Strong record of connecting model outputs to operating metrics and go/no-go decisions.

DECISION SCIENCE CAPABILITIES

Experimentation & Causal Measurement: Bayesian structural time series (BSTS), synthetic control, test/control measurement, pre/post analysis, incrementality, heterogeneous segment effects, A/B testing

Product & Marketplace Analytics: Supply-demand economics, pricing elasticity, conversion, contribution margin, fulfillment risk, segmentation, sales/visits/units/households

Modeling & Optimization: XGBoost, Random Forest, linear regression, forecasting, integer programming, optimization

Tools: Python, SQL, R, Spark, Snowflake, SageMaker, MLflow, Tableau

PROFESSIONAL EXPERIENCE

Data Science Lead | Hana E Solutions **Oct 2026–Present**

Remote, San Jose, CA

- **Agentic AI decision workflows:** Founding data science contributor building agentic AI decision workflows from scratch for two client engagements, starting with pricing; currently in proof-of-concept phase.

Senior Data Scientist | Vista **Oct 2021–Jul 2026**

San Jose, CA (Remote)

Marketplace Pricing & Decision Systems

- **Supply-demand equilibrium pricing:** Led two data scientists to replace static on-demand pricing bands with a supply-demand equilibrium engine. Framed off-fleet fulfillment as the primary risk metric; one 43-aircraft fleet saw 28% fewer off-fleet bookings in the first month after release (pre/post comparison).
- **Supply and demand modeling:** Modeled available supply through six submodels and expected demand through quote volume and price-sensitive conversion; used a bounded Brentq solver to set yield where expected demand aligned with supply, subject to operational floors and ceilings.
- **Regional pricing grid:** Launched a 7×7 regional pricing recommendation grid across seven U.S. regions with revenue-management overrides; tracked recommendation-to-override differences to diagnose adoption and where human judgment improved decisions.
- **Interpretable model design:** Chose interpretable supply/demand components over direct price-to-booking prediction because historical pricing choices were confounded by routes, aircraft, and discretionary manager behavior.
- **EU cost model and margin:** Developed a gradient-boosted cost prediction model for EU instant-booking prices using ~1,800 records; validated with shadow-mode invoices and historical backtests before relaunch. Margin improved from −8% to +17% and loss-making bookings fell from 55% to 20%.

Customer, Conversion & Product Workflows

- **Member booking forecasts:** Owned Program member booking forecasts feeding MSRP recommendations for 14 fleet categories; automated production scoring and daily yield/pricing reports. Improved forecast accuracy from 71% to 78% and reduced RMSE from 1.9 to 1.4.
- **Lead propensity:** Built behavioral XGBoost propensity scores (Hot/Warm/Cold) across 3/7/10-day lead activity windows, integrated into Salesforce to prioritize acquisition outreach and commercial follow-up.
- **Cancellation-policy badges:** Partnered with consumer product and engineering on LLM-derived cancellation-policy badges; structured refund terms across 532 operator contracts, with 84.7% badge-ready automatically and uncertain cases withheld for manual review.

Senior Data Scientist | 84.51° (Kroger) **Jul 2019–Oct 2021**

Cincinnati, OH

Causal Inference, Experimentation & Retail Optimization

- **BSTS causal measurement:** Designed and executed Bayesian structural time-series causal measurement for Quaker endcap merchandising in Richmond, comparing treatment performance with synthetic controls, general endcap benchmarks, and Quaker aisle products over the same ~3-month window.
- **Rollout decision:** Estimated 4% lift for Quaker endcaps versus 10% for the store-matched general-endcap benchmark (and 2% for aisle products). Presented sales, visits, and household measures and diagnostics; results informed the decision to pause a proposed Eastern-division rollout.
- **Battery endcap experiments:** Segmented stores by footprint and compared reduced-space stores with controls across sales, visits, units, and households; recommended selective space reductions only in segments without meaningful sales deterioration.
- **Bakery space optimization:** Led bakery space allocation across four categories: modeled expected sales with linear regression and Random Forest using 26 store features, then optimized limited footage with integer programming and category constraints. Nine pilot stores recorded 14% higher early sales than five historical comparison stores; treated result as directional due to cohort differences.

Data Scientist | Practo Technologies **Jun 2015–Apr 2018**

Bangalore, India

- **Reporting applications:** Built R/Shiny and Flask product and sales reporting applications, replacing an external analytics platform costing $4K/month per person.
- **Conversion analytics:** Applied customer-feedback NLP and product-journey A/B testing to identify conversion drop-offs and opportunities for intervention.

LEADERSHIP, EDUCATION & RECOGNITION

Leadership: Led two data scientists on pricing systems; founded cross-functional learning sessions for 60+ colleagues.

Patent: U.S. Patent No. 12,314,757 B1 (2025): Efficient resource scheduling using adaptive scheduling criteria (co-inventor).

Purdue University — M.S., Business Analytics & Data Science (2019)

BITS Pilani — B.E. (Hons.) and M.Sc., Applied Mathematics (2015)